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A dual reduction strategy for reduce-order modeling of periodic control system☆

Mohammad-Sahadet Hossain [a,*], Muhtasim Alam Chowdhury [b], Md. Samiur Rahman [b]

[a] Department of Mathematics and Physics, North South University, Dhaka, Bangladesh
[b] Department of Electrical and Computer Engineering, North South University, Dhaka, Bangladesh

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ABSTRACT

Model order reduction (MOR) of periodic systems using the Krylov subspace methods received lots of interest in last few decades. In this paper, a structured Krylov subspace based model reduction for linear discrete-time periodic (LDTP) control system has been proposed using the corresponding lifted form. The proposed model reduction strategy first finds a reduced order Lyapunov equation via a projection where the projection subspace is computed by using the periodic Arnoldi method. Then, by applying the balanced truncation model reduction approach, it forms a reduced order model of the original system. In both cases, the periodic structure is ensured in the solutions and systems' dynamics. This paper also presents numerical results to confirm the efficiency and accuracy of the proposed algorithm.

1. Introduction

Model reduction is an important tool to replace the large and complex dynamical system with a model of comprehensibly smaller size which is amenable to reproduce a similar system dynamics of the original large-scale model. In many engineering and mechanical problems, the linear discrete-time periodic (LDTP) systems representing those problems are very complex and hence very expensive for completely analyzing the original systems. Consequently, model order reduction (MOR) technique is used to overcome this problem. In this paper, a LDTP system is analyzed with the help of MOR approach where the Krylov subspace method and the balanced trucation method are exploited to generate the reduction frameworks for the original model.

The LDTP models are referred as the connecting bridge between the linear and nonlinear analysis of many mechanical and engineering problems. A linear discrete-time p-periodic system for $\tau = 0, 1, \dots, p-1$, and of order $\mathbf{n} = (n_0, n_1, \dots, n_{p-1})$ has the form

$$\begin{aligned} x_{\tau+1} &= A_\tau x_\tau + B_\tau u_\tau, \\ y_\tau &= C_\tau x_\tau, \qquad \tau \in \mathbb{Z}, \end{aligned} \tag{1}$$

where $A_\tau \in \mathbb{R}^{n_{\tau+1}\times n_\tau}$, $B_\tau \in \mathbb{R}^{n_{\tau+1}\times m_\tau}$, $C_\tau \in \mathbb{R}^{q_\tau \times n_\tau}$ are periodic with a period $p \geq 1$, and $\sum_{\tau=0}^{p-1} n_\tau = n$. In this paper, the system matrices are allowed to have the time-varying dimensions.

Throughout the last few decades, these kinds of systems have been investigated and studied in different literatures [1–6] in the fields of science and engineering. The most common components for such systems are the radio frequency (RF) circuits, where the systems are designed to exhibit near linear responses, but reveal strong nonlinearity for local oscillations. A very common approach

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* Corresponding author.
E-mail address: mohammad.hossain@northsouth.edu (M.-S. Hossain).

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to dump these nonlinear responses is the linearization of the nonlinear circuit models around a steady-state term. As a results the model is periodic and can be represented by a model similar to system (1). A state–space model like (1) can also be generated from multirate data sampling, attitude control dynamics of helicopter rotors, and the orbital motion modeling of spacecraft [7].

For system (1), a reduced order model (ROM) can be constructed as follows

$$\begin{aligned} \tilde{x}_{k+1} &= \tilde{A}_\tau \tilde{x}_\tau + \tilde{B}_\tau u_\tau, \\ \tilde{y}_\tau &= \tilde{C}_\tau \tilde{x}_\tau, \qquad \tau \in \mathbb{Z}, \end{aligned} \tag{2}$$

where the reduced dimension $\mathbf{r} = (r_0, r_1, \dots, r_{p-1})$. Also $\tilde{A}_\tau \in \mathbb{R}^{r_{\tau+1}\times r_\tau}$, $\tilde{B}_\tau \in \mathbb{R}^{r_{\tau+1}\times m_\tau}$, $\tilde{C}_\tau \in \mathbb{R}^{q_\tau \times r_\tau}$, $\sum_{\tau=0}^{p-1} r_\tau = r$, $r_\tau \ll n_\tau$ and $r \ll n$.

Model order reduction strategies are very useful tools where numerical simulations and investigations require a significant amount of computational cost for the original large model. In this MOR process, an efficient lower-order system is approximated from a higher-order dynamical system with maximum accuracy and lower computational cost. Model reduction for LDTP systems is now a days a topic of several state- of-art research including topics like control, spectral analysis and harmonic response, Helicopter rotor design, aerospace engineering and many more [2,8–15].

In the last few decades, balanced truncation method became very popular to find reduced order model of LDTP system (1) [13,16–18]. The main beauty of balanced truncation MOR approach is that it finds the solutions of the dual matrix equations, known as *periodic Lyapunov equations*, associated with the system (1). For the balanced truncation MOR of system (1), one requires to solve the periodic discrete-time Lyapunov equations (PDLEs)

$$A_\tau G_\tau A_\tau^T - G_{\tau+1} = -B_\tau B_\tau^T, \tag{3}$$

$$A_\tau^T P_{\tau+1} A_\tau - P_\tau = -C_\tau^T C_\tau, \tag{4}$$

where $G_{p+\tau} = G_\tau$, and $P_{p+\tau} = P_\tau$ for $\tau = 0, 1, \dots, p-1$.

In the past few years, attention has been devoted to the numerical solution large sparse Lyapunov equations using the ADI method [19,20], the Smith method [20–22], and Krylov subspace methods [23,24]. However, all these methods are proposed to the linear time-invariant (LTI) case. For the solutions of (3) and (4), direct methods have been proposed in [5,13] using the lifted representation of the periodic matrices. However, in certain conditions when the data is sparse and the model is large then direct approach is computationally expensive. Hence, in that case the iterative methods, like the Krylov subspace-based approximation is exploited. Krylov based approximation using the block Arnoldi algorithm have been proposed in [25] for standard periodic Lyapunov equations. Unfortunately, no detailed analysis and further implementation issues based on numerical results were found in any forthcoming paper of Daniel Kressner that he promised in [25]. Very recently, iterative methods, such as the Smith method [26], the ADI method [27], conjugate gradient methods [28], IRKA [29], BCR method [30] have been proposed to find low rank iterative solutions for large scale LDTP system (1). This paper extends the idea of [25] in the lifted representation of LDTP Lyapunov equations and discuss the implement issues of the periodic solutions to model reduction of periodic discrete-time systems.

This paper is outlined is as follows. In Section 2, the dynamics of the LDTP system and their Gramians have been reviewed briefly. Then, the periodic Lyapunov equations related to the LDTP system (1) are introduced. After that, the periodic reachability and observability Gramians from [5] are studied using the lifted representations. In Section 3, the Krylov subspace based numerical approach is discussed for generating the projection subspace. Section 4 represents the structure preserving periodic Arnoldi algorithm that generates the appropriate subspace for the reduced order model. In Sections Section 6, a demonstration on how the newly developed method can be applied to balanced truncation model reduction method is presented. Finally, in Section 7, numerical examples are presented to illustrate the efficiency and implementation of this model reduction technique. At the end of this paper, a conclusion is added in Section 8 and after that an acknowledgment is given.

2. The LDTP system: A detailed analysis

Generally the regularity and the eigenstructure of the LDTP system (1) is represented by it's state-transition matrix. Here, it can be defined as

$$\psi_A(j,i) = A_{j-1} A_{j-2} \cdots A_i, \quad j > i, \tag{5}$$

where $\Phi_A(i,i) = I_{n_i}$. The state-transition matrix over a period p beginning from time j, is called the Monodromy matrix at time j, is defined by

$$\psi_A(j+p,j) = A_{j+p-1} A_{j+p-2} \cdots A_j \in \mathbb{R}^{n_j \times n_j}. \tag{6}$$

The monodromy matrix is capable of holding many important characteristics of the LDTP system (1). A detailed analysis of these periodic characteristics as well as the concepts of observability, reachability and minimality of the periodic systems can be found in [4,12,13,26].

The eigenvalues of (6) are called *characteristic multipliers* at time j. If all eigenvalues (6) lie inside the open unit disk, then the periodic system (1) will be asymptotically stable [13]. The asymptotically stable system (1) is said to be *reachable* at time τ if the periodic reachability Gramian defined as

$$G_\tau = \sum_{i=-\infty}^{\tau-1} \psi_A(\tau, i+1) B_i B_i^T \psi_A(\tau, i+1)^T \in \mathbb{R}^{n_\tau \times n_\tau}, \tag{7}$$

is nonnegative, i.e., $G_\tau \geq 0$. It is *observable* at time τ if the periodic observability Gramian defined as

$$P_\tau = \sum_{i=\tau}^{\infty} \psi_A(i,\tau)^T C_i^T C_i \psi_A(i,\tau) \in \mathbb{R}^{n_\tau \times n_\tau}, \tag{8}$$

is nonnegative, i.e., $P_\tau \geq 0$. System (1) is *completely reachable* iff $G_\tau > 0$ for $\tau = 0, 1, \dots, p-1$, and *completely observable* iff $P_\tau > 0$ for $\tau = 0, 1, \dots, p-1$.

These periodic Gramians are then used to identify the stability analysis and many important applications, like the model order reduction of system (1). Note that these periodic Gramians are the solutions of the PDLEs (3) and (4). These types of periodic matrix equations also appear while solving periodic state-feedback problems of periodic discrete-time system [9,11,13,30–32].

2.1. Representation of lifted LDTP system

Lifted representation is a very important tool for analyzing and modeling of periodic discrete-systems [1,2,6]. In lifted representation, the periodic discrete-time system is represented as a LTI system where inputs, states and outputs of the original periodic system are placed in the cyclic places on a lifted system. Here, the *cyclic lifted representation* is considered from [4].

System (1) can be represented using the *cyclic lifted representation* as follows

$$\mathcal{X}_{\tau+1} = \mathcal{A}\mathcal{X}_\tau + \mathcal{B}\mathcal{U}_\tau, \quad \mathcal{Y}_\tau = \mathcal{C}\mathcal{X}_\tau, \tag{9}$$

where

$$\begin{aligned} \mathcal{B} &= \operatorname{diag}(B_0, B_1, \dots, B_{p-1}), \\ \mathcal{A} &= \begin{bmatrix} 0 & \cdots & 0 & A_0 \\ A_1 & & & 0 \\ & \ddots & & \vdots \\ 0 & & A_{p-1} & 0 \end{bmatrix}, \\ \mathcal{C} &= \begin{bmatrix} 0 & \cdots & 0 & C_0 \\ C_1 & & & 0 \\ & \ddots & & \vdots \\ 0 & & C_{p-1} & 0 \end{bmatrix}, \end{aligned} \tag{10}$$

The inputs, states and outputs of (9) are related to those of (1) as $\mathcal{X}_\tau = [x_1^T, \dots, x_{p-1}^T, x_0^T]^T$, $\mathcal{U}_\tau = [u_0^T, u_1^T, \dots, u_{p-1}^T]^T$, and $\mathcal{Y}_\tau = [y_0^T, y_1^T, \dots, y_{p-1}^T]^T$, respectively. Using this cyclic representation, the transfer function of (9) can be defined as

$$\mathcal{H}(z) = \mathcal{C}(zI_n - \mathcal{A})^{-1}\mathcal{B}. \tag{11}$$

The PDLEs (3) and (4) can be written as equivalent periodic lifted discrete-time algebraic Lyapunov equations (PLDLEs) by following the structure of the lifted system (9),

$$\mathcal{A}\mathcal{G}\mathcal{A}^T - \mathcal{G} = -\mathcal{B}\mathcal{B}^T, \tag{12}$$

$$\mathcal{A}^T\mathcal{P}\mathcal{A} - \mathcal{P} = -\mathcal{C}^T\mathcal{C}, \tag{13}$$

respectively, where $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ are as in (10), and the solutions are given by

$$\mathcal{G} = \operatorname{diag}(G_1, \dots, G_{p-1}, G_0), \quad \mathcal{P} = \operatorname{diag}(P_1, \dots, P_{p-1}, P_0).$$

The matrices $\mathcal{G}$ and $\mathcal{P}$ are *the lifted reachability and observability Gramians* of the lifted system (9), respectively. While solving the state-feedback problems and computing the gradients of optimal periodic problems such equations arise. Also such equations arise in the solutions of periodic Riccati equations [5,25], including applications to model order reduction.

3. The Krylov subspace approach

Krylov subspace based methods are well established for iterative approximation of the Lyapunov equations where the coefficient matrices are very large and sparse. The well known Arnoldi method [33] mainly constructs an orthonormal basis of the Krylov subspace. In this paper, a MIMO system is considered for the model reduction process. Consequently, the block Krylov subspace which is a generalization of the standard Krylov method has been in the key focus of our study.

The cyclic lifted coefficient matrix $\mathcal{A}$ and the block diagonal matrix $\mathcal{B}$ generate the block Krylov subspace given by

$$\mathcal{K}_l(\mathcal{A}, \mathcal{B}) = \operatorname{span}(\mathcal{B}, \mathcal{A}\mathcal{B}, \mathcal{A}^2\mathcal{B}, \dots, \mathcal{A}^{l-1}\mathcal{B}). \tag{14}$$

The block Krylov subspace method first computes the solution $\mathcal{R}$ of the reduced Lyapunov equation

$$(\mathcal{V}_l^T\mathcal{A}\mathcal{V}_l)\mathcal{R}(\mathcal{V}_l^T\mathcal{A}\mathcal{V}_l)^T - \mathcal{R} = -(\mathcal{V}_l^T\mathcal{B})(\mathcal{V}_l^T\mathcal{B})^T, \tag{15}$$

where $\mathcal{V}_l$ is an orthonormal basis of $\mathcal{K}_l(\mathcal{A},\mathcal{B})$. A *block Arnoldi process* can be used in that context which has the form

$$\mathcal{A}\mathcal{V}_l = \mathcal{V}_l\mathbf{H}_l + h_{m+1,m}v_{m+1}e_m^T = \mathcal{V}_{l+1}\hat{\mathbf{H}}_l,$$

where the matrix $\mathcal{V}_{l+1}$ contains the orthonormal columns of $\mathcal{V}_l$. This choice was originally justified by the fact that under certain condition on $\hat{\mathbf{H}}_l$ and for large l, such a decomposition process will generate the orthonormal basis $\mathcal{V}_l$ of the Krylov subspace.

Then, for sufficiently large l, $\mathcal{V}_l\mathcal{R}\mathcal{V}_l^T$ will be a good approximation of the solution of the unreduced Lyapunov equation (12), i.e., $\mathcal{G} \approx \mathcal{V}_l\mathcal{R}\mathcal{V}_l^T$. Considering $\beta = \|\mathcal{B}\|_F$, and e_1 as the first column of the identity matrix I_m, we can define the residual matrix R_m for (12) as follows

$$R_m = \mathcal{A}\mathcal{V}_l\mathcal{R}\mathcal{V}_l^T\mathcal{A}^T - \mathcal{V}_l\mathcal{R}\mathcal{V}_l^T + \beta^2\mathcal{V}_l e_1 e_1^T\mathcal{V}_l^T. \tag{16}$$

Here, the most crucial task is to choose an efficient $\mathcal{V}_l$ so that the block Arnoldi iterations preserve the block diagonal structure at each iteration step. It is very important to preserve such a block diagonal structure so that one can easily read off the periodic orthonormal basis $V_\tau^{[l]}$, $\tau = 0, 1, \ldots, p-1$, from the lth iteration $\mathcal{V}_l$, i.e.,

$$\mathcal{V}_l = \mathrm{diag}(V_1^{[l]}, \ldots, V_{p-1}^{[l]}, V_0^{[l]}), \tag{17}$$

for $l = 1, 2, 3, \ldots$, and so on. In the next section, details of this structure preserving block Arnoldi iteration steps and its implementation will be discussed.

4. Periodic block Arnoldi algorithm with structure preservation

By applying the Arnoldi process to the Krylov subspace $\mathcal{K}_l(\mathcal{A},\mathcal{B})$, we generate the matrix $\mathcal{V}_l$. The columns of matrix $\mathcal{V}_l$ form an orthonormal basis of $\mathcal{K}_l(\mathcal{A},\mathcal{B})$. Note that, while computing $\mathcal{K}_l(\mathcal{A},\mathcal{B})$, each block of the block Arnoldi process has a new cyclic structure, except the first one which is diagonal. The reason behind this is that, except the first block, each new block in the block Arnoldi process of $\mathcal{K}_l(\mathcal{A},\mathcal{B})$ is generated by a left multiplication of a cyclic block matrix $\mathcal{A}$ with its' previous block.

Consider $\mathcal{K}_l(\mathcal{A},\mathcal{B})$ in (14) as the block Krylov subspace. Clearly, right hand side's first block is a block diagonal matrix. The second block is a multiplication of a block cyclic matrix with a block diagonal matrix which creates a block cyclic matrix. The Arnoldi process is sought to preserve the block diagonal structure at each iteration step. Because of this, a new permutation matrix is introduced in each of the iteration steps of (14) to make each iteration a block diagonal matrix.

Consider the following block diagonal matrix

$$\Pi = \begin{bmatrix} K_{m_0} & 0 & \cdots & 0 & 0 \\ \vdots & K_{m_1} & 0 & & 0 \\ & & \ddots & & \vdots \\ 0 & & & K_{m_{p-2}} & 0 \\ 0 & 0 & \cdots & & K_{m_{p-1}} \end{bmatrix}, \tag{18}$$

where K_{m_i} is a square identity matrix of size m_i, and m_i = column rank(B_i).

Then a permutation matrix $\alpha^i\Pi$ is introduced at each block iteration step i in the computation of (14). Note that, at each iteration step i, $\alpha^i\Pi$ will change in a cyclic manner by a forward block-column shift. Then, for $i = 1$, the permutation matrix is as follows

$$\alpha^1\Pi = \alpha\Pi = \begin{bmatrix} 0 & K_{m_0} & \cdots & 0 & 0 \\ \vdots & 0 & K_{m_1} & & 0 \\ & & \ddots & & \vdots \\ 0 & & & 0 & K_{m_{p-2}} \\ K_{m_{p-1}} & 0 & \cdots & & 0 \end{bmatrix}; \tag{19}$$

Similarly, for $i = 2$, the newly generated permutation matrix is as follows

$$\alpha^2\Pi = \begin{bmatrix} 0 & 0 & K_{m_0} & 0\cdots & 0 \\ \vdots & \vdots & 0 & K_{m_1} & 0 \\ & & \vdots & \vdots & * \\ K_{m_{p-2}} & & & 0 & 0 \\ 0 & K_{m_{p-1}} & 0 & 0\cdots & 0 \end{bmatrix}; \tag{20}$$

For example, suppose $p = 3$, and $\tau = 0, 1, 2$. Then, for $i = 1$, the permutation matrix is as follows

$$\alpha^1\Pi = \alpha\Pi = \alpha(\Pi) = \begin{bmatrix} 0 & K_{m_0} & 0 \\ 0 & 0 & K_{m_1} \\ K_{m_2} & 0 & 0 \end{bmatrix},$$

which can be referred to as forward shift of the last block-column of Π in (22). Similarly, when $i = 2$, the newly generated permutation matrix will be

$$\alpha^2\Pi = \alpha(\alpha\Pi) = \alpha(\alpha(\Pi)) = \begin{bmatrix} 0 & 0 & K_{m_0} \\ K_{m_1} & 0 & 0 \\ 0 & K_{m_2} & 0 \end{bmatrix}. \tag{21}$$

This newly generated permutation matrix is a forward shift of the last block-column of $\alpha\Pi$. For the solutions of the periodic Lyapunov equations, a similar type of computation is also proposed in [25]. The permutation matrix $\alpha^i\Pi$ satisfies the relations $\alpha^p\Pi = \alpha^0\Pi = \Pi$, where p is the period of the system (1), and precisely

$$\alpha^p\Pi = \Pi = \begin{bmatrix} K_{m_0} & 0 & 0 \\ 0 & K_{m_1} & 0 \\ 0 & 0 & K_{m_2} \end{bmatrix}. \tag{22}$$

is a forward shift of the last block-column of $\alpha^2\Pi$. Now moving on to the second block in (14), where the product matrix $\mathcal{AB}$ is a block cyclic matrix. By a left multiplication with a permutation matrix, a block diagonal matrix $\mathcal{AB}(\alpha\Pi)$ is reformulated. One thing to be noted that the sizes of the block identity matrices in Π will depend on the sizes of the corresponding blocks of product matrices in $\mathcal{AB}$. Similarly, the third block, i.e., $\mathcal{A}^2\mathcal{B}$ will have a block diagonal structure after left multiplication with the permutation matrix $\alpha^2\Pi$.

As long as the desired order of the Krylov approximation or the desired convergence is not achieved, similar kind of process will continue. Hence, the new form of (14) is as follows

$$\mathcal{K}_l(\mathcal{A}, \mathcal{B}) = \text{span}(\mathcal{B}, \mathcal{AB}(\alpha\Pi), \mathcal{A}^2\mathcal{B}(\alpha^2\Pi), \dots, \mathcal{A}^{l-1}\mathcal{B}(\alpha^{l-1}\Pi)). \tag{23}$$

Algorithm 1 presents the whole algorithm of the Krylov approximation.

Algorithm 1 Periodic Arnoldi algorithm

Input: $\mathcal{A}$, $\mathcal{B} = \text{diag}(B_0, B_1, \dots, B_{p-1})$
Output: Orthonormal basis $\mathcal{V}_l$

1: **for** $\tau = 0, 1, \dots, p-1$ **do**
2: Compute the QR decomposition $V_\tau^1 F_\tau^1 = \text{QR}(B_\tau)$,
3: $V_{\tau+1}^{[1]} = V_\tau^1$
4: **end for**
5: Construct $\mathcal{V}^1 = \text{diag}(V_1^{[1]}, \dots, , V_{p-1}^{[1]}, V_0^{[1]})$ % $V_p^{[1]} = V_0^{[1]}$
6: $\mathcal{V}_l = \mathcal{V}^1$
7: **for** $j = 1, 2, 3, \dots,$ **do**
8: $\mathcal{W}_{j+1} = \mathcal{A}\mathcal{V}^j$
9: $\mathbf{W} = \mathcal{W}_{j+1}(\alpha^j\Pi)$
10: **for** $i = 1$ to j **do**
11: $\mathbf{H}_{i,j} = (\mathcal{V}^i)^T\mathbf{W}$
12: $\mathbf{W} = \mathbf{W} - \mathcal{V}^i\mathbf{H}_{i,j}$
13: **end for**
14: Compute the QR decomposition $\mathcal{V}^{j+1}\mathbf{H}_{j+1,j} = \text{QR}(\mathbf{W})$
15: $\mathcal{V}_l = [\, \mathcal{V}_l \quad \mathcal{V}^{j+1} \,]$
16: **end for**

In Algorithm 1, due to the QR decomposition, the block diagonal structure in $\mathbf{W}$ will be destroyed. For preserving the block diagonal structure in $\mathbf{W}$, and hence, in $\mathcal{V}_l$, the QR decomposition is used on the factored form of $\mathbf{W}$, where $\mathbf{W} = \text{diag}(W_{1,i}^b, \dots, W_{p-1,i}^b, W_{0,i}^b)$ at every iteration step i. This means that the QR decomposition is computed in Step 14 of Algorithm 1 as

$$[\hat{V}_i, \hat{Q}_i, r_{cc}] = \text{QR}((W_{k,i}^b)^T, \xi_{cc}),$$

where $\xi_{cc} = \sqrt{\epsilon}$ [29], and update

$$W_{\tau,i}^b = \hat{Q}_i\hat{V}_i^T[\, I_{r_{cc}}, 0\,]^T,$$

for $\tau = 0, 1, \dots, p-1$. After that $\mathcal{V}^j = \text{diag}(\hat{V}_{1,i}^b, \dots, \hat{V}_{p-1,i}^b, \hat{V}_{0,i}^b)$ is constructed at every iteration step i, and then it is used in the $i+1$-th iteration step. This ensures the block diagonal $\mathcal{V}_l$ in step (15) at each $j = 1, 2, \dots,$ to the last iteration step.

This is also important that the columns of $\mathcal{V}^{j+1}$ are orthogonal to each other and orthogonal to the previously computed columns in $\mathcal{V}_l$. It is recommended to use the rank-revealing QR decomposition [34] here that reduces the memory requirements by truncating the redundant columns of $\mathcal{V}^j$ and also guarantee that $\mathcal{V}_l$ is an orthonormal basis of $\mathcal{K}_l(\mathcal{A}, \mathcal{B})$.

However, a modified version of Algorithm 1 is proposed in this paper where the periodic coefficient matrices and their corresponding lifted forms are used simultaneously to generate the orthonormal basis of $\mathcal{K}_l(\mathcal{A}, \mathcal{B})$. To ensure this orthonormality, here some mathematical manipulations need to be exploited. In Algorithm 2, the modified iterative computation of the basis $\mathcal{V}_l$ is presented.

Algorithm 2 Modified version of Algorithm 1

Input: $\mathcal{A}$, $\mathcal{B} = \mathrm{diag}(B_0, B_1, \ldots, B_{p-1})$,Tolerance $= \varepsilon$.
Output: Orthonormal basis $\mathcal{V}_l$.

```
for τ = 0, 1, ..., p − 1 do
    Compute the QR decomposition, [q, F_τ] = QR(B_τ)
    V(:, :, τ + 1) = q/ ‖ q ‖
end for
Construct W_diag = diag(V(:, :, 1), ..., V(:, :, p − 1), V(:, :, 0))
for j = 1, 2, 3, ..., iter do
    V_l = [ ]
    W = A W_diag
    W_diag = W(σ^j Π),        % W_diag = diag(W(:, :, 1), ..., W(:, :, p − 1), W(:, :, 0))
    Identify W(:, :, i) from W_diag for i = 0 : p − 1
    for m = 0, 1, ..., p − 1 do
        Ŵ = W(:, :, m),   V̂ = V(:, :, m)
        for s = 1, ..., n_k do
            for i = 1, ..., r_k do
                h = V̂(:, i)^T Ŵ(:, s)
                Ŵ(:, s) = Ŵ(:, s) − h V̂(:, i)
                if s > 1 then
                    for s̃ = 1 : s − 1 do
                        h̃ = Ŵ(:, s̃)^T Ŵ(:, s)
                        Ŵ(:, s) = Ŵ(:, s) − h̃ Ŵ(:, s̃)
                    end for
                end if
            end for
            if ‖Ŵ(:, s)‖ < ε then
                for ρ = s : n_k − 1 do
                    Ŵ(:, ρ) = Ŵ(:, ρ + 1)
                end for
                n_k = n_k − 1
                Ŵ = Ŵ(:, 1 : n_k)
            else
                Ŵ(:, s) = Ŵ(:, s)/‖Ŵ(:, s)‖
            end if
        end for
        V̂ = [V̂, Ŵ]
        V_l = diag[V_l, V̂]
        if size(Ŵ, 2) = 0 then
            return
        end if
        W(:, :, m) = Ŵ
    end for
    Construct W_diag = diag(W(:, :, 1), ..., ..., W(:, :, p − 1), W(:, :, 0))
end for
```

$$(\mathcal{V}_l)^T \mathcal{A} \mathcal{V}_l = \begin{bmatrix} 0 & & \cdots & 0 & H_0^l \\ H_1^l & & & & 0 \\ & \ddots & & & \vdots \\ 0 & & & H_{p-1}^l & 0 \end{bmatrix} =: \hat{\mathcal{A}}_l,$$

$$(\mathcal{V}_l)^T \mathcal{B} = \mathrm{diag}(F_1^1, \ldots, F_{p-1}^1, F_0^1) =: \hat{\mathcal{B}}_l,$$

where the matrices F_τ^1 for $\tau = 0, 1, \ldots, p-1$, are computed within the first loop of Algorithm 1. As a result of this, the reduced Lyapunov equation (15) is as follows

$$\hat{\mathcal{A}}_l \, \mathcal{R} (\hat{\mathcal{A}}_l)^T - \mathcal{R} = -\hat{\mathcal{B}} \hat{\mathcal{B}}_l^T, \tag{24}$$

and the solution $\mathcal{R} = \mathrm{diag}(\hat{G}_1, \ldots, \hat{G}_{p-1}, \hat{G}_0)$ preserves the block diagonal structure as in (12), and $\hat{G}_\tau$ are the approximate solutions of (3) for $\tau = 0, 1, \ldots, p-1$. Also a similar type of approximation can be achieved for the other PLDLEs in (13).

5. Cyclic computations for solutions of PLDLEs

In fact, the Krylov iterations implemented in Algorithm 1 not only ensure that each particular block in (23) is block diagonal at each iteration step l, but facilitate also to compute the periodic orthonormal basis $V_\tau^{[l]}$, from (23) directly for different τ.

From simple algebraic manipulations of (23), it is observed that the periodic matrices A_τ and B_τ appear in a cyclic manner in periodic computations of $V_\tau^{[l]}$ in every iteration step l. By observing these cyclic relations and handling them technically, the periodic Krylov subspaces $V_\tau^{[l]}$, $\tau = 0, 1, \dots, p-1$, $l = 1, 2, \dots$, can be computed directly. Some of those computations are given below.

For $l = 1$, the computations appear as follows

$$\begin{aligned}
V_0^{[1]} &= V_{p-1}^1, \text{where } B_{p-1} = V_0^{p-1} R_0^{p-1},\\
V_1^{[1]} &= V_0^1, \text{where } B_0 = V_0^1 R_0^1,\\
V_2^{[1]} &= V_1^1, \text{where } B_1 = V_1^1 R_1^1,\\
&\vdots\\
V_{p-1}^{[1]} &= V_{p-2}^1, \text{where } B_{p-2} = V_{p-2}^1 R_{p-2}^1.
\end{aligned}$$

For $l = 2$, the computations are

$$\begin{aligned}
V_0^{[2]} &= A_{p-1} V_{p-1}^{[1]},\\
V_1^{[2]} &= A_p V_p^{[1]} = A_0 V_0^{[1]},\\
V_2^{[2]} &= A_1 V_1^{[1]},\\
&\vdots\\
V_{p-1}^{[2]} &= A_{p-2} V_{p-2}^{[1]}.
\end{aligned}$$

For $l = 3$, we have

$$\begin{aligned}
V_0^{[3]} &= A_{p-1} A_{p-2} V_{p-2}^{[1]} = A_{p-1} V_{p-1}^{[2]},\\
V_1^{[3]} &= A_p A_{p-1} V_{p-1}^{[1]} = A_0 V_0^{[2]},\\
V_2^{[3]} &= A_1 A_p V_p^{[1]} = A_1 A_0 V_0^{[1]} = A_1 V_1^{[2]},\\
&\vdots\\
V_{p-1}^{[3]} &= A_{p-2} A_{p-3} V_{p-3}^{[1]} = A_{p-2} V_{p-2}^{[2]},
\end{aligned}$$

and so on. Therefore, the first block $V_1^{[l]}$ of the block diagonal $\mathcal{V}^l$ is generated from the Krylov subspace

$$\mathcal{K}_1^{[l]}(A_\tau, V_\tau) = \text{span}(V_1^{[1]}, A_0 V_0^{[1]}, A_0 V_0^{[2]}, \dots, A_0 V_0^{[l-1]}). \tag{25}$$

The second block $V_2^{[l]}$ will be generated from the Krylov subspace

$$\mathcal{K}_2^{[l]}(A_\tau, V_\tau) = \text{span}(V_2^{[1]}, A_1 V_1^{[1]}, A_1 V_1^{[2]}, \dots, A_1 V_1^{[l-1]}). \tag{26}$$

Similarly, the other blocks of $\mathcal{V}^l$ are generated. Finally we get the last block $V_p^{[l]} = V_0^{[l]}$ from

$$\mathcal{K}_0^{[l]}(A_\tau, V_\tau) = \text{span}(V_0^{[1]}, A_{p-1} V_{p-1}^{[1]}, A_{p-1} V_{p-1}^{[2]}, \dots, A_{p-1} V_{p-1}^{[l-1]}). \tag{27}$$

Considering the appearance of the periodic matrices in different iterations l, and for different period $\tau = 0, 1, \dots, p-1$, we summarize it in Algorithm 3.

As we see from Algorithm 3 that the iterations can gather redundant information if Algorithm 3 is executed for larger iteration steps. In that case the rank-revealing QR decomposition [34] is recommended to eliminate those redundancies and to guarantee $\mathcal{V}^l$ an orthonormal basis of $\mathcal{K}_l(\mathcal{A}, \mathcal{B})$.

6. The balanced truncation MOR

Balanced truncation model reduction means that one finds a balanced realization of the periodic system (1) by using appropriate projection onto the original system and then truncates the states that do not contribute significantly to the dynamics of the system. This projection matrices are constructed from the product of the periodic Gramian factors.

Assume that, after completion of successful iteration steps using Algorithm 2 (or, equivalently Algorithm 3), $G_\tau \approx \hat{G}_\tau$ is achieved. It is also possible to achieve a similar approximation of the solutions of (4). Assume the Cholesky factorizations of the periodic Gramians to be $G_\tau = R_\tau R_\tau^T$ and $P_\tau = M_\tau M_\tau^T$. For a balanced realization, the singular value decompositions (SVDs) for $\tau = 0, 1, \dots, p-1$, are computed as follows

$$M_\tau^T R_\tau = [U_{\tau,1}, U_{\tau,2}] \begin{bmatrix} \Sigma_{\tau,1} & \\ & \Sigma_{\tau,2} \end{bmatrix} [W_{\tau,1}, W_{\tau,2}]^T, \tag{28}$$

Algorithm 3 Periodic Arnoldi for cyclic computational solutions of PLDLEs

Input: $A_\tau \in \mathbb{R}^{n_{\tau+1}\times n_\tau}$, $B_\tau \in \mathbb{R}^{n_{\tau+1}\times m_\tau}$, $\tau = 0, 1, \dots, p-1$; and an integer $l \le n_\tau$.
Output: Matrices $V_\tau^{[l]}$, $\tau = 0, 1, \dots, p-1$, (l is iteration number),
so that $\mathcal{V}^l$=diag$(V_1^{[l]}, \dots, V_{p-1}^{[l]}, V_0^{[l]})$.

for $\tau = 0, 1, \dots, p-1$ **do**
  Compute the QR decomposition: $B_\tau = V_\tau^1 R_\tau^1$
  $V_{\tau+1}^{[1]} = V_\tau^1$
end for
for $l = 1, 2, \dots,$ **do**
  for $\tau = 0, 1, \dots, p-1$ **do**
    if $\tau = 0$ **then**
      $\tau' = p-1$
    else
      $\tau' = \tau - 1$
    end if
    $Z_\tau^l = (V_\tau^{[l]})^T A_{\tau'} V_{\tau'}^l$
    $L = A_{\tau'} V_{\tau'}^l - V_\tau^{[l]} Z_\tau^l$
    Compute the rank-revealing QR decomposition:
    $LM = V_\tau^{l+1} N$
    $V_\tau^{[l+1]} = [V_\tau^{[l]}, \quad V_\tau^{l+1}]$
  end for
end for

where $\Sigma_{\tau,1} = \text{diag}(\sigma_{\tau,1}, \dots, \sigma_{\tau,r_\tau})$, $\Sigma_{\tau,2} = \text{diag}(\sigma_{\tau,r_\tau+1}, \dots, \sigma_{\tau,n_\tau})$, with $\sigma_{\tau,1} > 0$. Here, these are the Hankel singular values (HSVs) of the system. Also $U_{\tau,1}, W_{\tau,1} \in \mathbb{R}^{n_\tau \times r_\tau}$. From the decompositions above, the truncation matrices are defined as follows

$$S_{\tau,r} = M_\tau U_{\tau,1} \Sigma_{\tau,1}^{-1/2}, \quad T_{\tau,r} = R_\tau V_{\tau,1} \Sigma_{\tau,1}^{-1/2},$$

and also the reduced order system (2) of dimension r_τ, $\tau = 0, 1, \dots, p-1$, is computed as

$$\tilde{A}_\tau = S_{\tau+1,r}^T A_\tau T_{\tau,r}, \ \tilde{B}_\tau = S_{\tau+1,r}^T B_\tau, \ \tilde{C}_\tau = C_\tau T_{k,r}. \tag{29}$$

Though the original system is comprised of time constant dimensions, it is not compulsory to insist on the time constant dimensions for the reduced order system.

Now consider $\tilde{\mathcal{H}}(z)$ to be the transfer function of the reduced order lifted system computed using the reduced matrices of (29). Then the error in the reduced system is defined by the $\mathbb{H}_\infty$-error norm [13] as

$$\begin{aligned} \|\mathcal{H} - \tilde{\mathcal{H}}\|_{\mathbb{H}_\infty} &= \sup_{\omega\in[0,2\pi]} \|\mathcal{H}(e^{i\omega}) - \tilde{\mathcal{H}}(e^{i\omega})\|_2 \\ &\le 2\,\text{trace}\,(\text{diag}(\Sigma_{0,2}, \Sigma_{1,2}, \dots, \Sigma_{p-1,2})). \end{aligned} \tag{30}$$

The error bound of the system will bound the absolute error $\|\mathcal{H} - \tilde{\mathcal{H}}\|_2$ between the norm of the original system and the reduced system as defined in relation (30). This is one of the most important feature of balancing based MOR approach. In the next section numerical results from two different examples have been included which illustrate appropriateness of the MOR process proposed in this paper.

7. Simulation and numerical results

This section represents some outcomes of the numerical experiments to demonstrate the efficiencies and accuracy of the methods discussed so far.

7.1. Example 1

The first example is an artificial model problem where the periodic discrete-time descriptor system is reformulated from its main model in [14, Example 1]. In this reformulation, we consider

$$r_1 = 0.5, \quad r_2 = 0.05, \quad r_3 = -0.02, \quad r_4 = 0.12,$$

$$A_{01} = \begin{bmatrix} r_1\cos(\pi/3) & r_1\sin(\pi/3) \\ -r_1\sin(\pi/3) & r_1\cos(\pi/3) \end{bmatrix}, \quad A_{02} = \begin{bmatrix} r_2\cos(7\pi/5) & r_2\sin(7\pi/5) \\ -r_2\sin(7\pi/5) & r_2\cos(7\pi/5) \end{bmatrix},$$

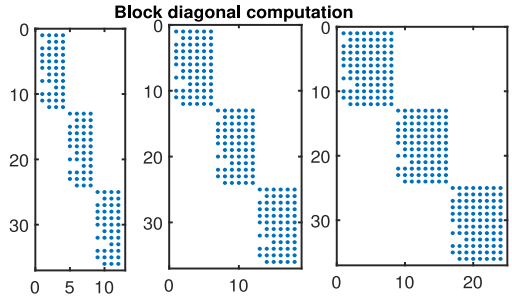

Fig. 1. The block diagonal computations of $\mathcal{V}_l$ at the 2nd, 3rd and 4th iteration steps (left to right, respectively).

$$A_{03} = \begin{bmatrix} r_3\cos(\pi/4) & r_3\sin(\pi/4) \\ -r_3\sin(\pi/4) & r_3\cos(\pi/4) \end{bmatrix}, \quad A_{04} = \begin{bmatrix} r_4\cos(\pi/10) & r_4\sin(\pi/10) \\ -r_4\sin(\pi/10) & r_4\cos(\pi/10) \end{bmatrix},$$

$$A = \operatorname{diag}(A_{01}, A_{02}, A_{03}, A_{04}).$$

Furthermore, for $\tau = 0, 1, 2$, we have

$$\theta_\tau = 2\pi\tau/p, \qquad p = 3,$$

$$w_\tau = \cos(\theta_\tau), \quad s_\tau = \sin(\theta_\tau).$$

and

$$B = \begin{bmatrix} 4 & -1 & 3 & 5 & 0 & -2 & 0 & 8 \\ 1 & 1 & s_\tau + 1 & -2 & 1 & 0 & 0 & -3 \end{bmatrix}^T,$$

$$C = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.05 + w_\tau \end{bmatrix}.$$

We then compute

$$A_d = 0.23A, \quad B_d = 0.05B, \quad C_d = 0.03C,$$

and finally compute the p-periodic system matrices as follows

$$\begin{aligned} A_\tau &= \operatorname{diag}(0.4I_2, 0.5\tau A_d, 0.4I_2); \qquad (I_2 \text{ is a} 2\times 2 \text{ identity matrix}) \\ B_\tau &= [\, I_2,\ (B_d)^T,\ 0.2I_2\,]^T; \\ C_\tau &= \left[\, C_d, [1,\ 0,\ 0]^T, [0,\ 1,\ 0]^T, [2,\ 0,\ 0]^T, [0,\ 2,\ 0]^T \,\right]. \end{aligned}$$

The reformulated periodic system has $n_\tau = 12$, $m_\tau = 2$ and $q_\tau = 3$ for $p = 3$ and $\tau = 0, 1, 2$. As a result the lifted system (9) has dimension $n = 36$.

All the eigenvalues of $\psi_A(p-1, 0)$ lie inside the unit circle, which represents the asymptotically stability of the system. The projected matrices were constructed using Algorithm 2 (or analogously using Algorithm 3). In Fig. 1, the block diagonal structure preserving computations of $\mathcal{V}_l$ are plotted using Algorithm 2. For the computation of $\mathcal{V}_l$, Algorithm 2 has been executed for 4 iteration steps.

In Fig. 2(a), all the HSVs of the periodic subsystems are presented for $\tau = 0, 1, 2$ that are computed by solving an equivalent reduced Lyapunov equation (24) and then computing the SVDs in relation (28). Fig. 2(b) presents the 24 HSVs (of largest magnitudes) from the 36 HSVs of the original system in lifted form. These are also the HSVs of the periodic subsystems but now in a descending order in terms of their magnitudes. We approximate system (1) to the tolerance 10^{-2} and truncate the states corresponding to the smallest 6 HSVs from these 24. We constructed a reduced order model using relation (29). Our reduced order model has the dimension $\mathbf{r} = (6, 6, 6)$ for $\tau = 0, 1, 2$. In Fig. 2(a) we plotted the HSVs of the subsystems of $\tau = 0, 1, 2$. The HSVs of the original and the reduced order system in their lifted forms are shown in Fig. 2(b). We recognize a significant matching between those HSVs.

To investigate the efficiency of the reduced-order system, we plot the frequency responses of the original and reduced order lifted systems in the frequency range $[0, 2\pi]$, π is in radians. We also depict the absolute error in these frequency responses to investigate the error between them. Fig. 3 shows the fruitful results of matching the frequency responses. The error between them is very negligible and below the error bound.

The main lifted system has 6 inputs and 9 outputs. Therefore, we check the output–input relations for each input and output of the original and the reduced systems in the frequency domain. For an efficient reduction, the frequency responses for the individual components of the lifted transfer functions $\mathcal{H}(e^{i\omega})$ and $\tilde{\mathcal{H}}(e^{i\omega})$ will match significantly. In the following (in Figs. 4, 5, and 6) we depicted the bode diagrams and the phase plots of $\mathcal{H}(e^{i\omega})$ and $\tilde{\mathcal{H}}(e^{i\omega})$ at different outputs from different inputs. Note that here $\mathcal{H}_{i,j}(e^{i\omega})$ indicates the response and phase at the jth output from the ith input.

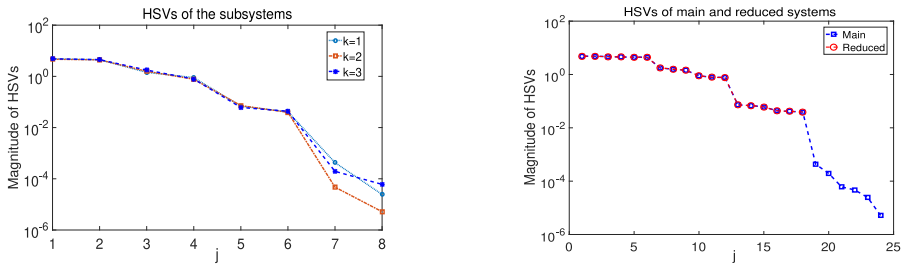

Fig. 2. (a) The HSVs of the subsystems (left), (b) The HSVs of the original (main) and reduced order systems in lifted form (right).

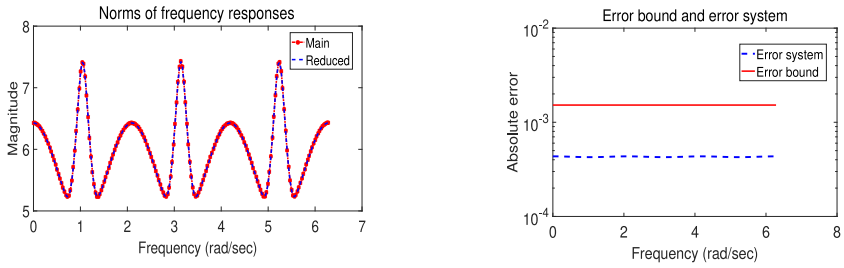

Fig. 3. The frequency responses of the original and the reduced-order lifted systems (left) and the error norms (right).

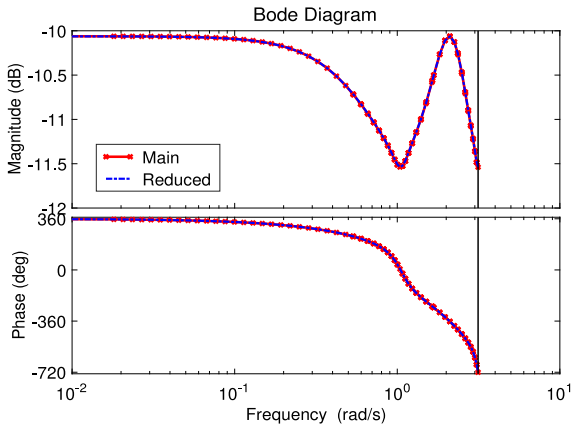

Fig. 4. Frequency responses for $\mathcal{H}_{1,1}(e^{i\omega})$ and $\tilde{\mathcal{H}}_{1,1}(e^{i\omega})$.

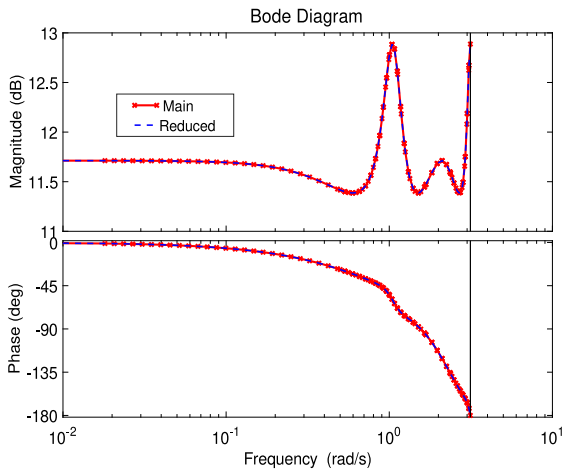

Fig. 5. Frequency responses for $\mathcal{H}_{2,5}(e^{i\omega})$ and $\tilde{\mathcal{H}}_{2,5}(e^{i\omega})$.

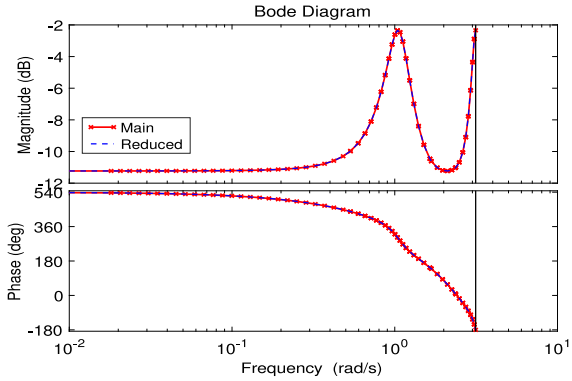

Fig. 6. Frequency responses for $\mathcal{H}_{6,1}(e^{i\omega})$ and $\tilde{\mathcal{H}}_{6,1}(e^{i\omega})$.

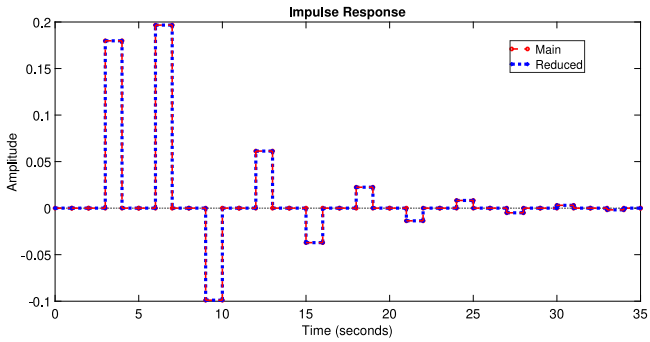

Fig. 7. Impulse responses of $\mathcal{H}$ and $\tilde{\mathcal{H}}$ from the 1st input to the 1st output channel.

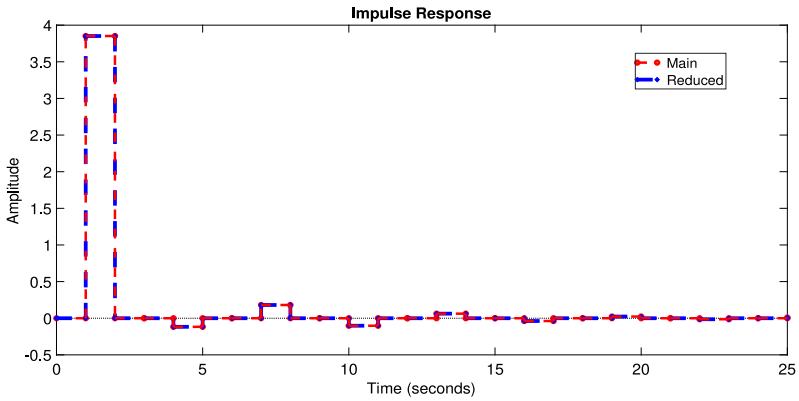

Fig. 8. Impulse responses of $\mathcal{H}$ and $\tilde{\mathcal{H}}$ from the 2nd input to the 5th output channel.

In all these above figures, we observe satisfactory match between the frequency responses of the original and the reduced order system in lifted form.

The accuracy in implementing the proposed algorithms can be seen from the time-domain analysis of the original and reduced systems. Hence, we investigate the impulse responses generated from the transfer functions of the original and reduced order lifted systems for different input to output channels. For the lifted systems, we get a 6-by-9 array of plots showing comparative pictures of impulse responses of the original and reduced lifted systems. Below in Figs. 7 and 8, we depict two such comparative pictures.

Here we find a significance match between the impulse responses. Similar satisfactory results are also observed in other comparative pictures of impulse responses of the original and reduced lifted systems.

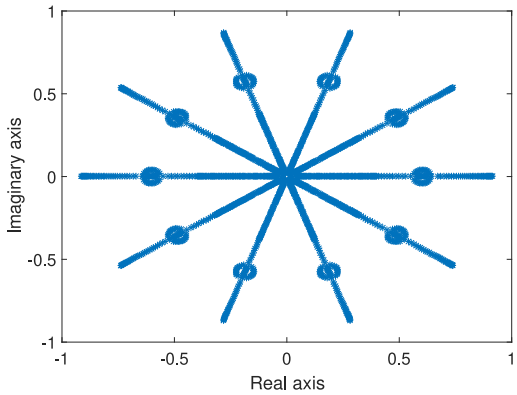

Fig. 9. Eigen-structure of the original (main) lifted system.

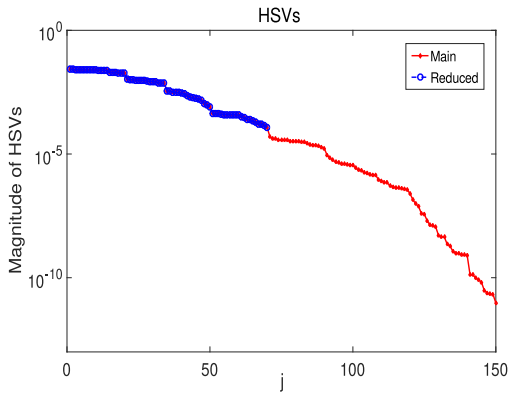

Fig. 10. The HSVs of the original (main) and the reduced-order lifted systems.

7.2. Example 2

As a second model problem we consider here a discrete-time periodic model from [26]. This periodic model was generated form an artificial continuous time piezo-mechanical systems described in Section 4.3 of [35]. The original model formulation is given at the appendix of [26]. For our data generation, we consider $n = 400$, $l = 20$, $nin = 2$, $nout = 3$ in the description of [26]. We also consider $E = I$, an identity matrix of compatible order . Hence the dimension of the continuous-time model is $2n + l = 820$. Therefore, the converted periodic discrete-time model with period $p = 10$, we have $n_\tau = 820$, $m_\tau = 2$, $p_\tau = 3$, and the order of the lifted system is $n = 8200$.

The system is asymptotically stable. This stability pattern based on eigenvalues is shown in Fig. 9. Fig. 10 presents the 150 HSVs (of largest magnitudes) from the 8200 HSVs of the original system in lifted form. For this approximation Algorithm 2 is executed 5 times. We approximate system (1) to the tolerance 10^{-8} and constructed a reduced order system of $\mathbf{r} = (13, 13, 13, 12, 13, 13, 13, 14, 12, 13)$ for $\tau = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9$, respectively.

Similar to Example 1, we plot the frequency responses for $\mathcal{H}(e^{i\omega})$ and $\tilde{\mathcal{H}}(e^{i\omega})$ in Fig. 11(a) the frequency range $[0, 2\pi]$. The error plot is also shown in Fig. 11(b). We observe similar satisfactory results for the original and reduced order system in lifted form.

8. Conclusion

In this paper the model reduction problem is discussed for periodic discrete-time system using the structured Krylov approximation. The method first construct an orthonormal basis for the reduction of the periodic Lyapunov equation. The ROM generated from this proposed algorithm shows very efficient and accurate results in terms of Gramians' factor approximation, and also in time and frequency responses. However, the paper has some limitations as well. The algorithms need to be generalized for descriptor system where $E_\tau \neq I$ and E_τ can be singular for any value of τ.

We also need to investigate other similar MOR approach, specially the IRKA [29] and the ADI method [27] to approximate the solutions of the periodic Lyapunov equations. This will ensure us to produce a comparative picture of the MOR strategies for the LDTP systems. Working with real life large sparse data will be the future extension of this current research work. To investigate the suitability of parallel computation for the algorithms will be a attractive task for future extension of this paper.

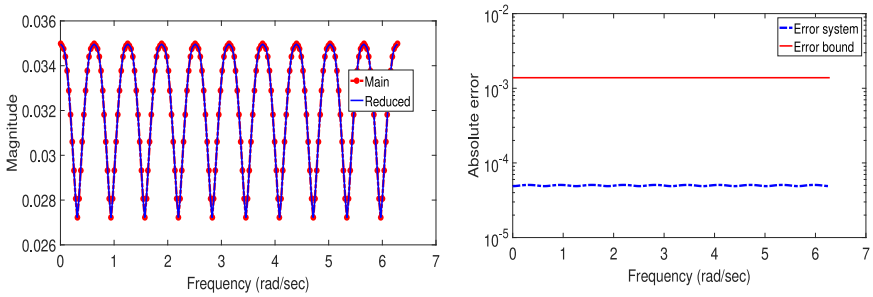

Fig. 11. (a) The frequency responses (left) and (b) the error in the frequency responses (right) of the original (main) and the reduced-order lifted systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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